

Satyabrata Das

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EDUCATION

University of Florida

M.S., Artificial Intelligence Systems

Gainesville, FL

Aug 2025 – May 2027

Coursework: Machine Learning for AI, Applied Deep Learning, AI Systems, AI Safety & Robustness, Image Processing & Computer Vision

EXPERIENCE

Evvy

Computer Vision Engineer – Eye Tracking & ML

Florida, USA (Remote, Startup)

May 2026 – Present

- Built real-time eye tracking and blink detection systems for AI interview monitoring using Python, OpenCV, and MediaPipe; improved gaze estimation accuracy and reduced false positives across varied lighting, webcams, and user environments.
- Prototyped PyTorch and TensorFlow models that classified user behavior from gaze and blink data and implemented anomaly detection, enabling the system to flag suspicious interview patterns with higher reliability.
- Optimized PyTorch inference pipelines to reduce latency, integrated browser-based computer vision using WebGL, and worked with founders to launch the solution into a deployment supporting multiple concurrent users.

ARC Document Solutions

Software Engineer (Full-time)

India

Jul 2022 – Sep 2024

- Engineered real-time WebSocket sync layer for ARC Facilities (20K+ MAU), cutting collaboration latency by 25% and reducing support tickets by ~30% quarter-over-quarter.
- Optimized ARC Print by integrating 5 payment gateways (Razorpay, Paytm, Cashfree, Braintree, Elavon) and reCAPTCHA, reducing failed transactions by 15%.
- Applied multithreading and Core Data to build responsive, high-performance apps with version control (Git), unit testing, and CI/CD practices in an agile environment.

SCI-BI Software Pvt Ltd

Software Trainee

India

Jul 2021 – Dec 2021

- Developed ETL and automated data-processing pipelines handling millions of records daily with 99%+ accuracy.
- Optimized SQL workflows and configured Bash scripts with Prometheus and Grafana monitoring, improving data availability and reducing processing latency for large-scale datasets.

SHIPPED PROJECTS

SummarIQ – NLP Pipeline for Document Intelligence

GitHub

- Built end-to-end NLP pipeline using HuggingFace Transformers (T5/BART) and LangChain for automated classification and semantic retrieval over 200+ documents, improving pipeline accuracy by ~18%.
- Deployed via FastAPI at sub-1.4s latency with FAISS-based chunking and embedding strategies for scalable semantic search and modular CI/CD-ready architecture.

Intelligent Bistro – ML-Powered Workflow Automation

GitHub

- Designed agentic AI system using Claude with structured tool calls to automate multi-step administrative workflows, achieving 95%+ task completion rate without human intervention.
- Reduced administrative processing time by ~60% through LLM-driven pipelines with schema validation, error handling, fallback mechanisms, and production observability logging.

JuggleIQ – Scalable Production ML Service

GitHub

- Architected scalable ML inference service with FastAPI + Docker on AWS, handling 1,000+ predictions per session at p95 latency under 200ms with full CI/CD and zero-downtime deployments.
- Built production monitoring with drift detection and quality validation, enabling proactive identification of model degradation; designed auto-scaling architecture for variable traffic patterns.

PoseInsight – Intelligent Movement Analysis System

GitHub

- Built end-to-end AI movement analysis platform using MediaPipe BlazePose and an LSTM-based temporal quality predictor to classify exercise form (good/bad squat, push-up) across 33 biomechanical landmarks.
- Benchmarked Random Forest and MLP baselines against LSTM for sequence-aware movement quality; integrated Gemini AI to deliver personalized coaching feedback and injury risk alerts via Streamlit.

TECHNICAL SKILLS

AI & ML: PyTorch, TensorFlow Lite, HuggingFace Transformers, Scikit-learn, LLMs, NLP, CUDA

Computer Vision: OpenCV, MediaPipe, YOLO, 3D Pose Estimation, Eye Tracking, Gaze Estimation, Video Analytics

Generative AI & Retrieval: RAG, Vector Databases, Embedding Models, Retrieval Systems

Programming: Python, C++, Swift, SQL

Systems & Tools: FastAPI, Docker, REST APIs, CI/CD, Git, Linux, Prometheus/Grafana, Weights & Biases